IM0802 – Responsible Artificial Intelligence
Assignment 2: “Paper Title Goes Here”

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Abstract

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1 Introduction 350

Israel, which allocates 8.8% of its GDP to defense [1], invests heavily in the development of autonomous weapon systems (AWS), with their Prime Minister announcing a $100 billion investment in homegrown arms production last December [2]. Israel’s most well-known AWS is the Iron Dome air defense system, which intercepts incoming projectiles in real time. Building on this foundation in artificial intelligence (AI)-assisted defense technology, Israel has expanded into offensive targeting applications such as “Habsora”.

Habsora ("The Gospel") is an Israeli Defense Forces (IDF) AI-targeting system developed primarily by Unit 8200 within the Military Intelligence Directorate. First deployed during the May 2021 Operation "Guardian of the Walls", it generates target recommendations using a multimodal intelligence pipeline that encompasses signal intelligence, telephone metadata, satellite and drone imagery, social graph analysis, and life-pattern modelling [3, 4]. The system is not a single model, but an ensemble of multiple specialised algorithms fused together [4]. During the 2023-2024 Gaza conflict, Habsora was reported to generate over one hundred target recommendations per day, which in prior years required approximately twelve months of human analysts [3, 5].

Habsora’s AI capability is the probabilistic classification and prioritisation of locations and individuals as military objectives. It assigns confidence scores derived from large-scale pattern matching across a shared intelligence pool, rather than positive identification in the traditional intelligence analysis [6]. Military commands nominally authorise each recommendation, yet their review window documented for each target was approximately twenty seconds [3].

This creates a stakeholder dynamic with both ethical and legal significance. It’s output influencing both IDF commands, who operate under high cognitive load and time pressure; Palestinian civilians in the target environment bearing the lethal consequences and the broader international community which depends on International Humanitarian Law (IHL) compliance for the legitimacy of armed conflict. Steen et al. [7] offer a framework for this tension. They propose a two-axis model of Meaningful Human Control (MHC) which distinguishes between high automation and genuine human control, which both require adequate information and real freedom of action. The central argument of this report is that Habsora structurally undermines the second condition, and that this gap carries cascading implications across the technical, sociotechnical, and governance layers identified by Verdiesen et al. [8] in their Comprehensive Human Oversight Framework (CHOF).

2 Ethical Issue Identification 750

The CHOF structures oversight across three layers: governance, socio-technical layer, and technical. Each of these layers is further analysed across three temporal phases: before, during, and after deployment of the autonomous weapon system. The governance layer concerns the legal and ethical conditions, investigating whether societal values and ethical principles are respected. The socio-technical layer focuses on the stakeholders involved, both the ones operating the system and those personally affected by it. The technical layer looks at the system itself, addressing issues such as automation bias and the system working as a black box, meaning the inner workings of the system are not transparent to users or stakeholders.

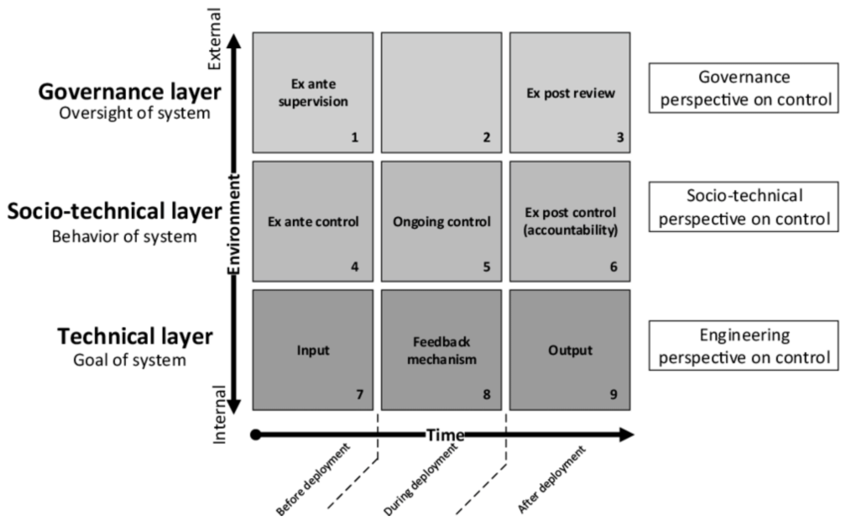

Figure 1: The CHOF framework for structuring ethical analysis across governance, socio-technical, and technical layers.

Underpinning all three layers is the Glass Box approach described by Aler Tubella et al. [9]. This approach translates abstract values, such as human dignity, into observable norms and requirements to be implemented in the the AI system. The Glass Box approach demands that the system's design and operation be transparent and verifiable, allowing for accountability and ensuring that ethical principles are upheld in practice. This is particularly relevant in the context of the IDF's use of the Lavender system, where the opacity of the AI's target recommendations raises significant ethical concerns regarding accountability and meaningful human control.

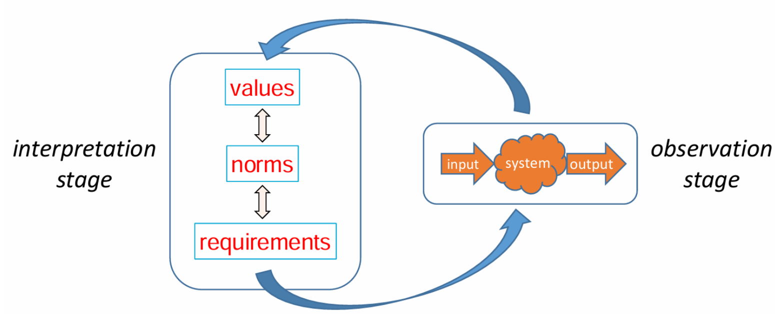

Figure 2: The two stages of the Glass-Box Approach: an Interpretation stage, where values are translated into design requirements, and an Observation stage, where we can observe and qualify the behaviour of the system

2.1 Governance Layer

2.1.1 Before

Article 36 of ‘Protocol I to the Geneva Conventions’ requires countries to conduct legal reviews of new weapons International Committee of the Red Cross [10]. Specifically to determine whether its employment would, in some or all circumstances, be prohibited by international law. These legal reviews examine whether the system can distinguish civilians from combatants, whether it can be used in a way that limits unnecessary suffering, and whether meaningful human control is retained. This article, and the rest of Protocol I, apply to countries that have ratified (formally accepted) it. Israel is one of the few countries that have not, and therefore is not legally obligated to conduct such reviews. There is a clear governance gap, because there is no legal requirement for the IDF to assess the compliance of Habsora with international humanitarian law (IHL) before deployment.

2.1.2 During

Under International Humanitarian Law (IHL), human commanders remain legally responsible for how weapons are used, even if the system has autonomous functions.In reality, commanders are judged on what they reasonably knew at the time of the decision, not hindsight. This creates accountability gaps when an AI system’s recommendation leads to civilian casualties. Additionally, in a high-tempo conflict environment, where decisions must be made rapidly, the real-time use of the system can not be effectively monitored, further complicating accountability.

2.1.3 After

Post-deployment, there is often an absence of meaningful review processes to assess the system’s compliance with IHL. It is difficult to attribute responsibility for any violations that may occur, as it may not be clear whether a particular recommendation was generated by the AI or if a human operator overrode it. This lack of transparency hinders accountability and undermines trust in the use of such systems in armed conflict.

2.2 Socio-technical Layer

2.2.1 Before

The affected stakeholders, Palestinian civilians, were likely not involved in the original design of Habsora, and their values were not elicited to ground the system’s targeting logic. The values and needs of those most impacted by the system were not adequately considered.

2.2.2 During

IDF commanders operating under time pressure may be susceptible to automation bias, where they over-rely on the system’s recommendations without sufficient critical evaluation. This can lead to an erosion of genuine human control, as defined by Steen et al. [7]. The system’s speed and opacity may make it difficult for operators to meaningfully review recommendations, further undermining human control over lethal decisions.

2.2.3 After

Palestinian civilians are non-consenting stakeholders who bear the consequences of the system’s use without any recourse. There is a significant asymmetry between those operating the system and those affected by it, raising ethical concerns about justice and fairness in the deployment of such technologies in armed conflict.

2.3 Technical Layer

2.3.1 Before

There were no verifiable, observable requirements anchoring the system’s recommendations, and the underlying model was opaque. This lack of transparency and accountability in the design phase raises significant ethical concerns about the system's compliance with IHL and its potential impact on civilians.

2.3.2 During

The speed of output generation is incompatible with meaningful human review, and there is a lack of explainability in individual targeting decisions. This technical opacity undermines the ability of operators to critically evaluate recommendations and maintain meaningful human control over lethal decisions.

2.3.3 After

The absence of audit trails makes post-hoc verification of norm compliance impossible. This lack of accountability and transparency in the system's operation raises significant ethical concerns.

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